

Notes: Exley & Kessler (QJE, 2022)

The Gender Gap in Self-Promotion

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1 Big picture

- **Research question.** Do men and women describe their own performance differently when they are equally performing, and can such subjective descriptions affect labor-market outcomes?
- **Core empirical object.** The paper studies *self-evaluations*: subjective answers to questions like “I performed well on the test,” willingness to apply for a job requiring similar skills, and expected success in such a job.
- **Why this matters.** Many labor-market situations ask people to describe their own skill using subjective language: resumes, interviews, promotion reviews, grant applications, presentations, and workplace meetings. If equally performing women use less favorable language, subjective evaluation processes can create unequal outcomes even without differences in actual ability.
- **Main setting.** Participants take a math-and-science test, a deliberately male-typed task. They then describe their performance using four self-evaluation questions.
- **Main result.** Women evaluate and promote their own performance substantially less favorably than equally performing men. In the baseline self-promotion setting, the female coefficient is about -13 points on a 0-100 performance scale.
- **Important design feature.** The paper compares equally performing men and women using performance fixed effects. This means the core estimates compare participants with the same number of correct answers.
- **Not just inaccurate beliefs.** The gap persists after participants are given perfect information about their absolute and relative performance. Beliefs about performance shrink the gap, but do not eliminate it.
- **Not just strategic promotion.** The gap persists when answers are private and not shared with employers. Therefore, the self-promotion gap appears to reflect an underlying self-evaluation gap rather than men responding more aggressively to promotion incentives.
- **Not general harshness.** The gap is small or absent when participants evaluate another person’s performance and when they evaluate themselves on a verbal task.
- **Early emergence.** In a sample of more than 10,000 middle-school and high-school students, girls provide less favorable evaluations than boys with the same measured performance. The gap appears as early as grade 6.
- **Economic consequences.** Employers respond positively to more favorable self-evaluations. Because women give lower self-evaluations, the paper shows that women can be less likely to be hired and earn lower expected wages than equally performing men.
- **Takeaway.** Gender gaps in subjective descriptions of performance can operate as a distinct channel through which labor-market gender gaps arise and persist.

2 Incremental contribution of the paper

- **From objective confidence to subjective self-description.** Earlier work often studies gender differences in beliefs about absolute or relative performance, willingness to compete, negotiation, or leadership. This paper isolates a different channel: how people *subjectively describe* performance even when performance can be objectively measured.
- **Clean measurement of self-promotion.** The paper designs self-evaluation questions that are subjective but quantitative. This makes self-promotion measurable, comparable across people, and usable in regressions.
- **Equal-performance comparisons.** The design compares women and men with the same test score, so the gender gap is not a mechanical reflection of actual performance differences.

- **Separating self-promotion from underlying self-evaluation.** By comparing public employer-facing self-evaluations with private self-evaluations, the paper shows that the gap is not simply a result of different strategic responses to incentives to inflate.
- **Separating beliefs from descriptions.** By providing perfect information about absolute and relative performance, the paper distinguishes inaccurate beliefs from subjective descriptions conditional on known performance.
- **Boundary testing across environments.** The paper tests social-norm information, immediate performance information, other-evaluation, a verbal task, and youth samples. This identifies where the gap is robust and where it disappears.
- **Link to real allocation decisions.** The employer experiment shows that subjective self-evaluations affect hiring and pay decisions. This connects the behavioral gap to economically meaningful outcomes.
- **Developmental evidence.** The youth sample documents that the gap exists in school-aged individuals, suggesting that the mechanism may emerge long before labor-market entry.
- **Policy implication.** The paper shifts attention away from only “fixing women” through confidence training and toward changing systems that rely heavily on subjective self-promotion.

3 Data, experimental design, and measurement

3.1 Participants and study waves

The authors run a sequence of online experiments and a youth study. The adult sample includes over 4,000 participants from MTurk and Prolific across five waves. The youth sample includes more than 10,000 students from grades 6 to 12.

- Wave 1 includes the Self-Promotion, Self-Promotion (Risky), and Private versions.
- Wave 2 replicates the Private version and introduces the Private (Social Norms) version.
- Wave 3 replicates the Private version and introduces the Private (Immediately Informed) version.
- Wave 4 replicates the Private (Immediately Informed) version and introduces the Private (Other-Evaluation) version.
- Wave 5 replicates the Private version and introduces the Private (Verbal) version.
- The youth sample uses a modified private design and asks comparable evaluation questions to students in grades 6 through 12.

Interpretation: The design is cumulative. Rather than relying on one experiment, the paper builds a sequence of versions that first documents the gap, then removes potential explanations one by one.

3.2 Task performance

In the main adult versions, participants complete a 20-question math-and-science test. The task is male-typed by design because the motivating labor-market gaps are especially pronounced in male-typed fields such as finance, technology, engineering, and computer science.

- The objective performance measure is the number of correctly answered questions out of 20.
- The analysis uses performance fixed effects, allowing comparisons of participants with the same exact score.
- The youth version uses a 10-question test and uses performance fixed effects for each possible youth score.

Interpretation: The performance fixed effects are central. Without them, gender differences in self-evaluation could be confounded by differences in test performance. With them, the coefficient on Female is the gap among equally performing participants.

3.3 The four evaluation questions

Participants answer four subjective evaluation questions.

- **Performance question:** agreement with a statement like “I performed well on the test” on a 0-100 scale.
- **Performance bucket question:** a six-point scale ranging from terrible to excellent performance.
- **Willingness-to-apply question:** willingness to apply for a job requiring performance on the same task.
- **Success question:** expected success in such a job.

Interpretation: These questions capture different manifestations of self-evaluation: a direct assessment, a categorical assessment, a labor-market application implication, and a success expectation. The gap appears across all four.

3.4 Self-Promotion version

In the Self-Promotion version, participants know that one self-evaluation answer may be shown to potential employers, who will use it to decide whether to hire them and how much to pay them.

- This creates incentives to describe performance favorably.
- The paper interprets responses in this setting as self-promotion.
- Figure II and Table II show that women provide lower self-promotion responses than equally performing men.

Interpretation: This is the paper’s closest analogue to labor-market communication, where applicants or employees often must describe their own performance subjectively.

3.5 Private version

In the Private version, self-evaluations are not shared with employers and do not affect payoffs.

- The version removes incentives to inflate answers.
- It also removes risk over employer payoffs, preferences over employer payoffs, and lack of control over how an employer interprets the answer.
- The gender gap remains large in Table II Panel C and Table III Panel C.

Interpretation: This is the key design step showing that the self-promotion gap is not just strategic behavior by men. The gap survives even when no one sees the answer.

3.6 Performance information and informed evaluations

Participants are later told their absolute and relative performance and answer the evaluation questions again.

- This tests whether gender differences in beliefs about performance explain the gap.

- Table III shows that informed gaps remain negative and statistically significant.
- The gap shrinks relative to uninformed evaluations but remains economically large.

Interpretation: Performance beliefs matter, but the paper’s central result is stronger: even when performance information is perfect, women subjectively evaluate the same performance less favorably than men.

3.7 Additional versions: mechanisms and boundaries

- **Private (Social Norms):** participants see average answers from prior participants with the same score. This tests whether uncertainty about appropriate subjective language drives the gap.
- **Private (Immediately Informed):** participants receive performance information before any self-evaluation. This removes anchoring or consistency motives from earlier uninformed answers.
- **Private (Other-Evaluation):** participants evaluate another person’s performance with the same score. This tests whether the gap reflects generally different standards.
- **Private (Verbal):** participants complete a verbal task. This tests whether the gap is specific to a male-typed domain.

Interpretation: The robustness versions are important because the paper is not merely documenting a gap. It is trying to locate the mechanism: the gap is about self-evaluation in a male-typed domain, not about general evaluation standards or strategic promotion alone.

4 Models and empirical specifications

4.1 Baseline regression

The main empirical specification is an OLS regression of the answer to a self-evaluation question on a female indicator and performance fixed effects:

$$Y_{iqv} = \alpha_q + \beta_q Female_i + \sum_s \gamma_{qs} \mathbf{1}\{Score_i = s\} + \varepsilon_{iqv}, \quad (1)$$

where Y_{iqv} is participant i ’s response to question q in study version v .

Interpretation: The coefficient β_q is the gender gap among participants with the same test score. Negative values mean women report less favorable self-evaluations than equally performing men.

4.2 Before and after performance information

The paper estimates the same structure before participants learn their score and after they receive perfect information:

$$Y_{iqv}^{pre} = \alpha_q^{pre} + \beta_q^{pre} Female_i + ScoreFE_i + u_{iqv}, \quad (2)$$

$$Y_{iqv}^{post} = \alpha_q^{post} + \beta_q^{post} Female_i + ScoreFE_i + e_{iqv}. \quad (3)$$

Interpretation: Comparing β^{pre} with β^{post} reveals how much of the gender gap is related to beliefs about performance. If the gap disappeared after information, beliefs would explain the result. It does not disappear.

4.3 Self-promotion versus private self-evaluation

The paper compares the employer-facing Self-Promotion version with the Private version. A conceptual pooled regression is:

$$Y_i = \alpha + \beta Female_i + \delta SelfPromotion_i + \theta(Female_i \times SelfPromotion_i) + ScoreFE_i + \varepsilon_i. \quad (4)$$

Interpretation: If men responded more strongly to promotion incentives, the interaction term would be negative and large. Instead, both men and women raise self-evaluations under promotion incentives, and the gender gap remains similar. This supports the interpretation that the gap is an underlying self-evaluation gap.

4.4 Youth grade-by-grade specification

For youth, the authors run OLS regressions separately by grade:

$$Y_{ig} = \alpha_g + \beta_g Female_i + \sum_s \gamma_{gs} \mathbf{1}\{Score_i = s\} + \varepsilon_{ig}. \quad (5)$$

Interpretation: Table IV shows that β_g is negative across grades 6-12. This means the gender gap emerges before workers enter the labor market.

4.5 Employer response and economic consequences

The employer task relates employer choices to workers' self-evaluation answers:

$$Hire_{jw} = \alpha + \rho SelfEvaluation_w + \eta_{jw}. \quad (6)$$

The paper reports that employers are more likely to hire workers who provide more favorable self-evaluations. Because women provide less favorable self-evaluations conditional on performance, this generates lower hiring probabilities and expected wages for equally performing women.

Interpretation: This result connects subjective self-description to allocation. The gender gap is not merely expressive; it can change who is hired and how much workers earn.

5 Identification and empirical design

5.1 Core identification problem

The main empirical challenge is distinguishing a gender gap in subjective descriptions from gender differences in actual performance, beliefs, strategic incentives, or standards.

- Actual performance is controlled using exact performance fixed effects.
- Beliefs are addressed by giving perfect performance information.
- Strategic incentives are addressed by comparing public self-promotion to private self-evaluation.

- Standards are addressed by asking participants to evaluate others.
- Domain specificity is addressed by comparing a math-and-science task to a verbal task.

Interpretation: The identification strategy is not one randomized treatment; it is a design sequence that systematically removes alternative explanations.

5.2 Why performance fixed effects are powerful

Performance fixed effects compare people with the same number correct. This controls nonparametrically for actual performance.

- The paper does not rely on linear controls for test score.
- It compares men and women at each score level.
- This matters because subjective evaluation may be nonlinear in performance.

Interpretation: A participant who scored 10 out of 20 is compared to others who also scored 10 out of 20. This makes the estimated gender gap a descriptive-language gap, not a score gap.

5.3 Why the Private version matters

The Private version removes the possibility that gender differences come from employer-facing incentives.

- If men inflate more strategically, the gap should shrink or vanish when answers are private.
- The gap remains about as large.
- Therefore, the employer-facing gap reflects a deeper self-evaluation difference.

Interpretation: This is the most important mechanism test. It pushes against the explanation that women simply promote less because they dislike strategic exaggeration.

5.4 Why the informed version matters

The informed version directly controls beliefs by design.

- Participants know their absolute and relative performance.
- The gap shrinks but remains significant.
- The remaining gap cannot be due to women merely thinking they scored lower.

Interpretation: The result reveals a distinction between knowing a fact and choosing subjective language to describe that fact. Two people can know the same performance and still describe it differently.

5.5 Why other-evaluation and verbal-task tests matter

The Private (Other-Evaluation) version and Private (Verbal) version define the boundaries of the result.

- Other-evaluation shows the gap is not simply a female tendency to evaluate performance more harshly in general.
- The verbal task shows the gap is not universal across all domains.
- Together, these tests point to self-evaluation in male-typed domains as the relevant mechanism.

Interpretation: The paper’s conclusion is narrower and stronger than a broad claim about women’s confidence. The gap depends on both the self and the task domain.

5.6 What the design cannot fully prove

- The experiments are online and lab-like, so external validity to high-stakes labor markets requires caution.
- The paper studies selected tasks, not every possible male-typed or female-typed domain.
- The youth evidence shows early emergence but does not identify the exact age or causal origin of the gap.
- The employer task shows consequences of subjective answers, but real employers often observe many other signals.

Interpretation: The paper is strongest as a causal and diagnostic design for isolating a mechanism. It does not claim to fully explain all labor-market gender gaps.

6 Section-by-section study guide

6.1 Introduction

The introduction motivates the paper with labor-market settings where people must describe their own performance using subjective language. It emphasizes that such descriptions can matter for hiring, pay, promotion, and perceived expertise.

6.2 Design, data collection, and setting

This section describes the experimental versions and the sequencing of waves. It is the key section for understanding how the paper separates self-promotion, beliefs, strategic incentives, social norms, evaluation standards, domain specificity, and age.

6.3 Results

The results section first documents a gender gap in employer-facing self-promotion, then shows the gap persists in private settings. It then shows that the gap is small or absent when evaluating others and absent on a verbal task.

6.4 Discussion

The discussion collects robustness, heterogeneity, and economic-consequence evidence. The employer and predictor results are especially important because they show that the gap can affect hiring and pay and may not be corrected by observers.

6.5 Youth evidence

The youth section shows that the gap appears across grades 6-12. This rules out a purely adult labor-market origin and suggests that the mechanism may be shaped by earlier socialization or stereotypes.

7 Tables and figures

7.1 Table I: Study Versions by Wave + Figure I: Performance and Absolute Performance Belief Distributions

Read:

- Table I maps the experimental architecture. Each wave either replicates a prior version or introduces a new version designed to test a mechanism.
- The Private version is repeatedly replicated across waves, making it the backbone of the robustness evidence.
- Figure I Panel A shows that women perform slightly better than men on the math-and-science test in the pooled adult data.
- Figure I Panel B shows that men report higher absolute performance beliefs than women.
- Figure I Panel C shows that the gender gap in beliefs is driven by women underestimating performance and men overestimating performance relative to actual performance.

Detailed interpretation: Table I shows that the paper’s design is cumulative rather than one-shot. The authors first establish the gap in self-promotion and private self-evaluation. Then, wave by wave, they remove candidate mechanisms: social norms, anchoring, other-evaluation, and verbal-domain generality. Figure I shows that the belief channel is real: before information, women believe they answered fewer questions correctly than men, even though performance is not lower. This motivates the informed-evaluation analyses in Table III.

Economic message: The first visual block establishes both the experimental map and a key mechanism: beliefs differ by gender, but the paper’s later results show that beliefs are not enough to explain the full self-evaluation gap.

TABLE I
STUDY VERSIONS BY WAVE

	Self-Promotion	Private	Self-Promotion (Risky)	Private (Social Norms)	Private (Imm. Informed)	Private (Other-Evaluation)	Private (Verbal)
Wave 1	New (N = 302)	New (N = 304)	New (N = 294)				
Wave 2		Replication (N = 302)		New (N = 298)			
Wave 3		Replication (N = 300)			New (N = 299)		
Wave 4					Replication (N = 597)	New (N = 597)	
Wave 5		Replication (N = 294)					New (N = 305)
Youth		Replication (N = 10,637)					

Notes. Data were collected in October 2018 for wave 1, November 2019 for wave 2, April 2020 for wave 3 and wave 4, and January 2021 for wave 5. Participants came from MTurk in waves 1–4 and from Prolific in wave 5. Youth data were collected in October 2020 as part of a partnership with the Character Lab Research Network, as described in Section V. In all but wave 4, we aimed to recruit 300 participants per study version. In wave 4, to generate more data from the *Private (Immediately Informed)* version, we aimed to recruit 600 participants per study version. Realized sample sizes for each study version appear in each cell.

(a) Table I: Study Versions by Wave

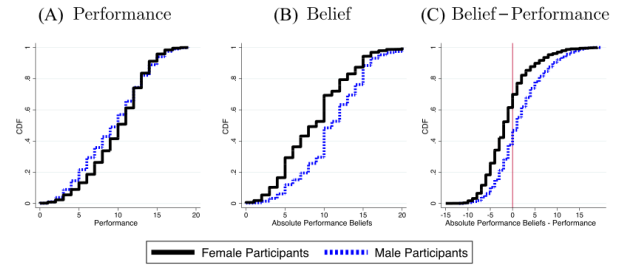


FIGURE I

Performance and Absolute Performance Belief Distributions

Graphs show CDFs for the associated outcome. Performance is the number of questions a participant answered correctly out of the 20 questions on the test. Belief is the number of questions a participant believes he or she answered correctly.

(b) Figure I: Performance and Absolute Performance Belief Distributions

7.2 Table II: Results from Evaluations Before Performance Information + Table III: Results from Informed Evaluations After Performance Information

Read:

- Table II shows that before performance information, the female coefficient is negative and statistically significant in every relevant math-and-science version.
- In the Self-Promotion version, women score about 12.7 points lower on the 0-100 performance question and about 15 points lower on willingness-to-apply and success questions.
- In the Private version, where responses are not shared with employers, the gap remains similarly large.
- Table III shows that after participants receive performance information, female coefficients remain negative and statistically significant.
- The informed gap is smaller than the uninformed gap, which means beliefs matter, but the informed gap remains economically meaningful.
- Panel I in Table III, the Other-Evaluation version, shows little or no gap when participants evaluate another person.
- Panel K in Tables II and III shows no significant gap on the verbal task.

Detailed interpretation: Tables II and III are the paper’s central result tables. Table II documents the raw self-evaluation gap conditional on performance. Table III asks whether the gap survives when participants know the truth about their own performance. The answer is yes. The comparison between Panels A-B and Panels C-F shows that employer-facing incentives are not necessary. The comparison with Panel I shows that the gap is not a general evaluation-standard gap. The comparison with Panel K shows that the gap is not present in the verbal domain.

Economic message: The tables show that the gap is robust, large, and not driven by either actual performance or ignorance of performance. The behavioral object is subjective interpretation and communication of the same performance.

TABLE II
RESULTS FROM EVALUATIONS (BEFORE PERFORMANCE INFORMATION IS PROVIDED)

Question:	Performance (1)	Performance bucket (2)	Willingness to apply (3)	Success (4)
Panel A: Self-Promotion version, wave 1 ($N = 302$)				
<i>Female</i>	-12.68*** (2.96)	-0.59*** (0.13)	-15.31*** (3.46)	-15.09*** (3.46)
Panel B: Self-Promotion (Risky) version, wave 1 ($N = 294$)				
<i>Female</i>	-9.15*** (2.93)	-0.47*** (0.13)	-12.82*** (3.29)	-9.24*** (3.32)
Panel C: Private version, wave 1 ($N = 304$)				
<i>Female</i>	-13.46*** (2.93)	-0.56*** (0.13)	-17.57*** (3.51)	-16.46*** (3.61)
Panel D: Private version, wave 2 ($N = 302$)				
<i>Female</i>	-12.21*** (3.18)	-0.55*** (0.15)	-17.25*** (3.54)	-14.39*** (3.53)
Panel E: Private (Social Norms) version, wave 2 ($N = 298$)				
<i>Female</i>	-15.14*** (3.28)	-0.80*** (0.16)	-16.93*** (3.71)	-15.62*** (3.71)
Panel F: Private version, wave 3 ($N = 300$)				
<i>Female</i>	-16.45*** (3.18)	-0.79*** (0.15)	-15.69*** (3.92)	-16.16*** (3.87)
Panel G: Private (Immediately Informed) version, wave 3: no evaluations				
Panel H: Private (Immediately Informed) version, wave 4: no evaluations				
Panel I: Private (Other-Evaluation) version, wave 4: no evaluations				
Panel J: Private version, wave 5 ($N = 294$)				
<i>Female</i>	-13.05*** (2.61)	-0.59*** (0.11)	-18.77*** (3.30)	-19.18*** (3.17)
Panel K: Private (Verbal) version, wave 5 ($N = 305$)				
<i>Female</i>	1.15 (2.40)	-0.12 (0.11)	1.99 (3.19)	-0.36 (3.02)
Panel L: All evaluations of own math and science performance ($N = 2,094$)				
<i>Female</i>	-13.83*** (1.13)	-0.67*** (0.05)	-17.28*** (1.31)	-16.12*** (1.32)
Performance fixed effects	Yes	Yes	Yes	Yes

Notes. * $p < .10$, ** $p < .05$, *** $p < .01$. Standard errors are robust. Results are from OLS regressions of the responses provided to the evaluation question noted in each column before participants are informed of their absolute and relative performance. Responses to the performance question indicate the extent of each participant's agreement (0-100) with the following statement: "I performed well on the test I took in part 1." Responses to the performance bucket question indicate which Likert-scale response (coded from 1 for the lowest to 6 for the highest) a participant selects when asked to "indicate how well you think you performed on the test in part 1?" Responses to the willingness to apply question indicate the extent of each participant's agreement (0-100) with the following statement: "I would apply for a job that required me to perform well on the test I took in part 1." Responses to the success question indicate the extent of each participant's agreement (0-100) with the following statement: "I would succeed in a job that required me to perform well on the test I took in part 1." *Female* is an indicator for the participant being female. Performance fixed effects are dummies

TABLE III
RESULTS FROM INFORMED EVALUATIONS (AFTER PERFORMANCE INFORMATION IS PROVIDED)

Question:	Performance	Performance bucket	Willingness to apply	Success
Panel A: Self-Promotion version, wave 1 ($N = 302$)				
<i>Female</i>	-7.01** (2.90)	-0.40*** (0.13)	-10.73*** (3.40)	-11.73*** (3.30)
Panel B: Self-Promotion (Risky) version, wave 1 ($N = 294$)				
<i>Female</i>	-7.24** (2.83)	-0.36*** (0.14)	-9.11*** (3.38)	-8.07** (3.29)
Panel C: Private version, wave 1 ($N = 304$)				
<i>Female</i>	-8.01*** (2.88)	-0.33** (0.14)	-13.25*** (3.53)	-13.15*** (3.53)
Panel D: Private version, wave 2 ($N = 302$)				
<i>Female</i>	-7.58** (3.18)	-0.42*** (0.15)	-14.15*** (3.53)	-14.37*** (3.46)
Panel E: Private (Social Norms) version, wave 2 ($N = 298$)				
<i>Female</i>	-11.93*** (3.15)	-0.62*** (0.16)	-16.39*** (3.42)	-15.77*** (3.58)
Panel F: Private version, wave 3 ($N = 300$)				
<i>Female</i>	-12.70*** (3.04)	-0.52*** (0.14)	-16.55*** (3.73)	-15.87*** (3.76)
Panel G: Private (Immediately Informed) version, wave 3 ($N = 299$)				
<i>Female</i>	-7.61** (3.35)	-0.47*** (0.16)	-11.42*** (3.81)	-12.48*** (3.61)
Panel H: Private (Immediately Informed) version, wave 4 ($N = 597$)				
<i>Female</i>	-8.54*** (2.22)	-0.42*** (0.10)	-16.63*** (2.42)	-18.66*** (2.30)
Panel I: Private (Other-Evaluation) version, wave 4 ($N = 597$)				
<i>Female</i>	0.29 (1.58)	-0.11 (0.08)	-3.54** (1.69)	-3.17* (1.68)
Panel J: Private version, wave 5 ($N = 294$)				
<i>Female</i>	-7.74*** (2.26)	-0.24** (0.10)	-12.91*** (3.09)	-14.24*** (3.01)
Panel K: Private (Verbal) version, wave 5 ($N = 305$)				
<i>Female</i>	-0.93 (1.94)	-0.05 (0.09)	-1.34 (2.76)	-1.36 (2.61)
Panel L: All evaluations of own math and science performance ($N = 2,990$)				
<i>Female</i>	-9.83*** (0.94)	-0.47*** (0.04)	-15.12*** (1.08)	-15.59*** (1.07)
Performance fixed effects	Yes	Yes	Yes	Yes

Notes. * $p < .10$, ** $p < .05$, *** $p < .01$. Standard errors are robust. Results are from OLS regressions of the responses provided to the question noted in each column, as defined in the notes of Table II, after participants are informed of their absolute and relative performance (or the other participant's absolute and

(a) Table II: Results from Evaluations Before Performance Information Is Provided

(b) Table III: Results from Informed Evaluations After Performance Information Is Provided

7.3 Figure II: Self-Promotion CDFs + Table IV: The Gender Gap in Evaluations Among Youth

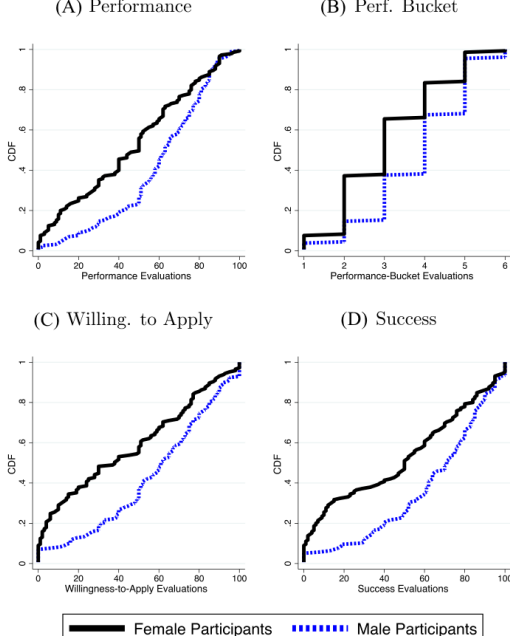
Read:

- Figure II plots CDFs of the four self-promotion answers in the baseline employer-facing version.
- For all four questions, the female distribution is shifted toward lower responses relative to the male distribution.
- The gap is visible not only in the mean but throughout much of the distribution.
- Table IV shows that among youth in grades 6-12, female students provide lower self-evaluations than male students with the same test score.
- The youth gap appears in performance, performance-bucket, willingness, and success questions.
- The gap persists after youth receive absolute performance information.

Detailed interpretation: Figure II makes the baseline gap visually transparent: for a given percentile, women's self-evaluation responses are lower. This matters because it rules out the idea that the result is driven by a small number of outliers. Table IV extends the mechanism to a younger population. The youth results are striking because the gap appears before many labor-market incentives are present. This suggests

the possibility of early socialization, stereotypes, or domain-specific norms about who is expected to be good at math and science.

Economic message: The baseline adult distribution and the youth evidence together show that the self-evaluation gap is broad, persistent, and potentially developmental. It is not only a labor-market strategic behavior.



(a) Figure II: In the Self-Promotion Version, CDFs Show the Gender Gap in Self-Promotion

TABLE IV
THE GENDER GAP IN EVALUATIONS AMONG YOUTH

	Among Students in Grade						
	6	7	8	9	10	11	12
Performance question							
Female	-10.52*** (1.26)	-11.81*** (1.04)	-11.05*** (0.79)	-11.80*** (1.45)	-12.14*** (1.41)	-11.40*** (1.49)	-10.44*** (1.74)
Performance bucket question							
Female	-0.47*** (0.07)	-0.56*** (0.05)	-0.51*** (0.04)	-0.59*** (0.07)	-0.52*** (0.07)	-0.53*** (0.08)	-0.45*** (0.09)
Willingness question							
Female	-6.82*** (1.60)	-6.48*** (1.31)	-3.68*** (1.00)	-3.86*** (1.82)	-6.92*** (1.88)	-0.29 (1.98)	-5.77** (2.38)
Success question							
Female	-9.42*** (1.52)	-9.85*** (1.24)	-7.19*** (0.93)	-7.41*** (1.73)	-8.40*** (1.76)	-4.58*** (1.69)	-7.29*** (2.16)
Informed performance question							
Female	-4.00*** (1.45)	-7.10*** (1.19)	-6.98*** (0.91)	-6.51*** (1.66)	-9.55*** (1.73)	-6.75*** (1.74)	-6.24*** (2.10)
Informed performance bucket question							
Female	-0.15** (0.07)	-0.33*** (0.06)	-0.27*** (0.05)	-0.27*** (0.09)	-0.33*** (0.09)	-0.26*** (0.09)	-0.22** (0.11)
Informed willingness question							
Female	-4.54*** (1.74)	-4.02*** (1.38)	-2.35** (1.03)	-3.43* (1.87)	-6.65*** (1.87)	0.00 (1.98)	-5.62** (2.39)
Informed success question							
Female	-5.02*** (1.68)	-7.42*** (1.36)	-4.94*** (1.01)	-4.61** (1.83)	-7.12*** (1.93)	-5.10*** (1.88)	-8.20*** (2.32)
N	1,521	2,208	3,367	1,031	989	871	650
Perf. fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes

(b) Table IV: The Gender Gap in Evaluations Among Youth

8 Result map

8.1 Result 1: Women self-promote less on the male-typed task

The Self-Promotion version shows large negative female coefficients across all four evaluation questions. Because employers see the answer, this setting captures self-promotion in a stylized labor-market environment.

8.2 Result 2: The gap is not fully explained by beliefs

Figure I shows women have lower absolute performance beliefs than men. However, Table III shows the gap remains after perfect performance information. Therefore, inaccurate beliefs are part of the story but not the whole story.

8.3 Result 3: The gap is not driven by strategic incentives

The Private version removes employer-facing incentives. The gap remains nearly as large as in the Self-Promotion version. This implies that the gap reflects underlying self-evaluations.

8.4 Result 4: Social-norm information does not close the gap

The Private (Social Norms) version tells participants how others with the same score answered. The gender gap remains. This suggests that uncertainty about typical or appropriate responses is not the main driver.

8.5 Result 5: Other-evaluation largely removes the gap

In the Other-Evaluation version, participants evaluate another person with the same score. The gender gap becomes small and often insignificant. This suggests the gap is about describing oneself, not about general standards for judging performance.

8.6 Result 6: The verbal task removes the gap

On the verbal task, female coefficients are small and statistically insignificant. This suggests the gap is domain-specific and especially relevant to male-typed tasks.

8.7 Result 7: Employer decisions translate the gap into economic consequences

Employers are more likely to hire and pay more to workers with more favorable self-evaluations. Since women give less favorable evaluations conditional on performance, the gap can generate worse outcomes for equally performing women.

8.8 Result 8: The gap emerges by middle school

Table IV shows negative female coefficients across grades 6-12. This evidence suggests that the source of the gap may predate labor-market entry.

9 Short summary

- The paper studies gender differences in subjective descriptions of performance.
- Participants complete a math-and-science test and then answer self-evaluation questions.
- Women perform at least as well as men in the pooled adult sample, but they report lower beliefs and lower self-evaluations.
- In employer-facing settings, women self-promote less than equally performing men.
- In private settings with no promotion incentives, the gender gap remains large.
- After perfect information about performance, the gap shrinks but remains statistically and economically significant.
- Social-norm information and immediate information do not eliminate the gap.
- Evaluating another person sharply reduces the gap.
- Evaluating performance on a verbal task eliminates the gap.
- Employer responses show that self-evaluations can affect hiring and pay.
- Youth data show that the gap appears as early as sixth grade.
- The paper argues that subjective self-description is a distinct mechanism that may contribute to gender gaps in labor-market outcomes.

10 Conclusion

10.1 Core conclusion

Exley and Kessler document a large gender gap in how equally performing men and women subjectively describe performance on a male-typed math-and-science task. The gap appears when evaluations are shared with employers and when they are private. It persists after performance information is provided and appears among youth as early as grade 6.

10.2 Economic intuition

The key distinction is between objective performance and subjective description. Two individuals can have the same score and the same information about that score, yet choose different words or numerical descriptions to characterize their performance. If institutions use subjective self-evaluations to allocate jobs, promotions, or pay, these subjective differences can have real consequences.

10.3 Why the evidence is persuasive

The paper’s strength is the sequence of design variations. Performance fixed effects compare equally performing participants. Informed evaluations address beliefs. Private versions remove promotion incentives. Social-norm treatments address uncertainty about appropriate responses. Other-evaluation and verbal-task versions define the boundary of the mechanism. Youth data show early emergence. Employer data show economic consequences.

10.4 Incremental lesson for labor-market inequality

The paper adds a channel to the study of gender gaps: not only differences in skill, confidence, negotiation, competition, or discrimination, but differences in the subjective framing of one’s own performance. This matters because many organizational processes ask people to advocate for themselves using imprecise language.

10.5 Policy implications

The results suggest that organizations should be cautious about evaluation systems that rely heavily on self-promotion, especially in male-typed domains. Possible responses include using more objective performance metrics, structured evaluation prompts, calibration information, evaluator training, and reducing the weight placed on self-assessed performance in hiring and promotion.

10.6 Limitations

The setting is stylized and online, and the tasks do not cover all labor-market domains. The paper does not identify the full developmental origin of the gap or all interventions that could close it. It also cannot fully capture how experienced real-world managers might combine self-evaluations with other information.

10.7 Takeaway

The central takeaway is that self-promotion gaps can arise even when performance is equal and even when people know their performance. Subjective self-evaluations are not neutral descriptions of objective facts; they are behavioral objects shaped by gender and domain, and they can influence economic outcomes.